

Assignment_Terraform_1

Terraform: State Locking and Workspaces

1. Why is state locking important? How does it prevent issues in a multi-user environment?

State locking is important because Terraform state is the source of truth for the infrastructure Terraform manages. The state file tells Terraform what already exists, what it created earlier, and what needs to change during the next plan or apply. If two people change the same state at the same time, Terraform can make decisions using outdated information.

In a multi-user environment, this can become risky very quickly. For example, one engineer may be applying a change while another engineer also starts an apply from their machine. Without locking, both operations could read the same old state, make different changes, and then try to write back to the state file. This may lead to corrupted state, duplicated resources, wrong deletions, or Terraform losing track of real infrastructure.

State locking prevents this by allowing only one Terraform operation to modify the state at a time. When one user is running a plan or apply that needs the state, Terraform locks it. Other users must wait until the lock is released. This keeps changes organized, avoids race conditions, and makes teamwork much safer.

2. What are workspaces in Terraform, and when would you use them?

Workspaces in Terraform are separate state environments inside the same Terraform configuration. Each workspace has its own state file, so the same set of Terraform files can track different versions of infrastructure without mixing their records together.

A common use of workspaces is to separate environments such as development, testing, staging, and production when their infrastructure is very similar. For example, the same Terraform code can be used for a dev environment and a production environment, but each workspace keeps its own state so Terraform knows which resources belong to which environment.

Workspaces are useful when the environments are mostly the same and only small values change, such as names, sizes, or environment labels. However, if environments are very different, it is usually better to use separate folders or separate configurations instead of forcing everything into workspaces.